RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019 - 2020 (Even Semester)

Innovative Practices Description

UNIT III 3G OVERVIEW

Degree, Semester & Branch: VI Semester B.E.ECE A & B
Course Code & Title: EC8004 Wireless Networks
Name of the Faculty member: Mrs.R.Ramalakshmi
Name of the Topic: UMTS Core network Architecture
Name of the Innovative Practice: Think pair share
Date & Duration: 07.02.2020 & 10 Minutes
ICT Tool Used: Smart Class Room

Description:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to think individually about a topic or answer to a question; and share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

Goals (Learning Outcomes):

- The students will be able to know about the concepts of UMTS network and reference architecture and about 3G technology

Use of appropriate methods:

Justification for choosing the topic using Think pair share activity:

Since the students are asked to think on the particular topic, they are able to think, remember and discuss the topics with their friends in the classroom. They are able to share their ideas on the particular topic by having discussions with them.

Effective presentation
Implementation(Plan & Execution) with Proof:

The students were in the class room. The topic was explained to the students first. Then they were asked to think about the topic and after sometime they were asked to pair themselves and to have a discussion on the particular topic..

Innovative Teaching Method Execution	
Think pair share on " UMTS Core network Architecture"	
	

<u>Significance of Results:</u>

<u>Assessment of Effectiveness/Success of the Activity:</u>

- ✓ The assessment of effectiveness of the activity was felt when the students were able to answer to the question on the particular topic when asked on the next week
- ✓ The students understood the topic very clearly since they had discussion with their classmates

<u>Reflective Critique:</u>

- **Benefits:**

  The students were more interested in learning the concepts if they think and discuss the topic and ask questions. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of UMTS network architecture by think pair share activity. They asked more doubts in the topic and learnt many things.

- **Challenges:**

  Basically some time is needed to explain the basic concepts of UMTS core network architecture. The slow learners found somewhat difficult in the initial stage later they also were able to think and share the concepts with their friends.

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO3	2	1	1		1					1		1

Changes required for future (if required):-

CO3: The students will be able to apply the suitable network depending on the availability and requirement

References:

1. Jochen Schiller, "Mobile Communications", Second Edition, Pearson Education 2012.
2. Vijay Garg , "Wireless Communications and networking", First Edition, Elsevier 2007.

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019 - 2020 (Even Semester)

Innovative Practices Description

UNIT III 3G OVERVIEW

Degree, Semester & Branch: VI Semester B.E.ECE A & B
Course Code & Title: EC8004 Wireless Networks
Name of the Faculty member: Mrs.R.Ramalakshmi
Name of the Topic: CDMA 2000 Overview
Name of the Innovative Practice: Animated Video
Date & Duration: 11.02.2020 & 10 Minutes
ICT Tool Used: Smart Class Room

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

- The students will be able to know about the concepts of CDMA 2000 and its radio network components

Use of appropriate methods:

Justification for choosing the topic using Animated Video activity:

Since the students visualize the concepts through video, they are interestingly watching the concepts and they are able to remember the concepts very easily. The students are able to answer the questions on the particular topic since they listened the video with more interest.

Effective presentation
Implementation(Plan & Execution) with Proof:

The animated video on the particular topic was played to the students in the class room. The students listened to the video with more interest and learnt the concepts of CDMA 2000 and its radio and network components.

Innovative Teaching Method Execution	
Animated video on “ CDMA 2000 Overview”	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students were able to answer to the question on the particular topic when asked on the next week
- ✓ The students understood the topic very clearly since they visualize the animated video

Reflective Critique:

- **Benefits:**

  The students were more interested in learning the concepts if they visualize the concepts. They were able to learn many things on the particular topic by watching the video

- **Challenges:**

  Basically some time is needed to play the animated video. The students need to watch the video for more time to understand the concepts fully. Later on they need to have a brief summary about the concept.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	1	1		1					1		1

Changes required for future (if required):-

CO3: The students will be able to apply the suitable network depending on the availability and requirement

References:

1. Jochen Schiller, “Mobile Communications”, Second Edition, Pearson Education 2012.
2. Vijay Garg , “Wireless Communications and networking”, First Edition, Elsevier 2007.